

Nash equilibrium

In a game, the best action for any given player depends, in general, on the other players' actions. So, when choosing an action, a player must have in mind the actions the other players will choose. That is, she must form a belief about the other players' actions. The assumption underlying is that each player's belief is **derived from her past experience playing the game**, and that this experience is sufficiently extensive that she knows how her opponents will behave.

Although we assume that each player has experience playing the game, we assume that she views each play of the game in isolation. She does not become familiar with the behavior of specific opponents and consequently does not condition her action on the opponent she faces; nor does she expect her current action to affect the other players' future behavior.

Assumptions:

- belief derived from experience playing the game
- play of the game in isolation
- not familiar with the behavior of specific opponents
- her action according to the model of rational choice, given her belief about the other players' actions.
- every player's belief about the other players' actions is **correct**.

A **Nash equilibrium** is an action profile $a^* = \{a_1^*, \dots, a_n^*\}$ with the property that no player i can do better by choosing an action different from a_i^* , given that every other player j adheres to a_j^*

Nash equilibrium of strategic game with ordinal preferences: The action profile $a^* = \{a_1^*, \dots, a_n^*\}$ in a strategic game with ordinal preferences is a Nash equilibrium if, for every player i and every action a_i of player i , a^* is at least as good according to player i 's preferences as the action profile $(a_i, a^*[-i])$ in which player i chooses $a_i \neq a_i^*$, while every other player j chooses a_j^* .

Equivalently, for every player i $u_i(a^*) \geq u_i(a_i, a^*[-i])$ for every action a_i of player i , where u_i is a payoff function that represents player i 's preferences.

- Nash equilibrium embodies a stable "social norm": if everyone else adheres to it, no individual wishes to deviate from it.
- The players know each others' preferences, and considers what each player can deduce about the other players' actions from their rationality and their knowledge of each other's rationality.

Let's see again our examples:

Prisoner's Dilemma

		Suspect 2	
		Quiet	Fink
Suspect 1	Quiet	2,2	0,3
	Fink	3,0	1,1

- (Quiet, Quiet) is not a Nash equilibrium because
 - if player 2 chooses Quiet, player 1's payoff to Fink exceeds her payoff to Quiet
 - if player 1 chooses Quiet, player 2's payoff to Fink exceeds her payoff to Quiet
- (Quiet, Fink) is not a Nash equilibrium because if player 2 finks it is better for player 1 to fink instead of being quite (finking she makes 3 years of prison while being quiet she stays for 4 years)
- (Fink, Quiet) is not a Nash equilibrium as the previous case
- (Fink, Fink) is a Nash equilibrium as if the other player finks the best is finking.

NOTICE: Nash equilibrium does not imply the best solution for all players.

Bach or Stravinsky?

		Player 2	
		Back	Stravinsky
Player 1	Back	2,1	0,0
	Stravinsky	0,0	1,2

- (Back, Stravinsky) and (Stravinsky, Back) are not Nash equilibria since it is better to agree with the other player and go together than split.
- (Back, Back) and (Stravinsky, Stravinsky) are Nash equilibria even if players have different opinion about which is better.

Coordination game: a variant of Bach or Stravinsky?

What if both the two players prefer going to Back instead of Stravinsky?

- As in the original game cooperation gives higher payoff
- Coordination would give best payoff

		Player 2	
		Back	Stravinsky
Player 1	Back	2,2	0,0
	Stravinsky	0,0	1,1

Again, however

- (Back, Stravinsky) and (Stravinsky, Back) are not Nash equilibria since it is better to agree with the other player and go together than split.
- (Back, Back) and (Stravinsky, Stravinsky) are **BOTH** Nash equilibria.

Is Stravinsky, Stravinsky possible? Yes, because in the case Stravinsky, Stravinsky still no player has interest to change its action.

Matching Pennies

		Player 2	
		Head	Tail
Player 1	Head	1, -1	-1, 1
	Tail	-1, 1	1, -1

- (Head, Head) and (Tail, Tail) are not Nash equilibria since in both cases it would be better for player 2 to change her choice
- (Head, Tail) and (Tail, Head) are not Nash equilibria since in both cases it would be better for player 1 to change her choice

This game has no Nash equilibria.

Stag Hunt

The table shows the case $n = 2$ but we need to consider every possible n

		Hunter 2	
		Stag	Hare
Hunter 1	Stag	2, 2	0, 1
	Hare	1, 0	1, 1

- (Stag, ..., Stag) is a Nash equilibrium because each player prefers this profile to that in which she alone chooses Hare.
- (Hare, ..., Hare) is a Nash equilibrium because each player prefers this profile to that in which she alone pursues the stag.
- No other profile is a Nash equilibrium

Strict vs nonstrict Nash equilibria

In all the Nash equilibria of the games a deviation by a player leads to an outcome worse for that player than the equilibrium outcome. The definition of Nash equilibrium, however, requires only that the outcome of a deviation be no better for the deviant than the equilibrium outcome. And, indeed, some games have equilibria in which a player is indifferent between her equilibrium action and some other action, given the other players' actions.

For a general game, an equilibrium is **strict** if each player's equilibrium action is better than all her other actions, given the other players' actions. Precisely, action profile a^* is a strict Nash equilibrium if for every player i : $u_i(a^*) > u_i(a_i, a^*[-i])$ for every action a_i of player i , where u_i is a payoff function that represents player i 's preferences.

The following game shows the implications of nonstrict equilibria

		player 2		
		L	M	R
player 1	T	1, 1	1, 0	0, 1
	B	1, 0	0, 1	1, 0

This game has a unique Nash equilibrium, namely (T, L). When player 2 chooses L, as she does in this equilibrium, player 1 is equally happy choosing T or B; if she deviates to B then she is no worse off than she is in the equilibrium. We say that the Nash equilibrium (T, L) is not a strict equilibrium.

Best response functions

When the number of players and/or the number of actions becomes large inspecting all the possible action profiles to find the Nash equilibria can be not feasible.

Consider a player, say player i . For any given actions of the players other than i , player i 's actions yield her various payoffs. We are interested in the **best actions**: namely the subset B of actions that yield her the highest payoff.

Formally B_i can be defined as the following function

$$B_i(a_{-i}) = \{a_i \in A_i : u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i}) \quad \forall a'_i \in A_i\}$$

Example:

In the prisoner dilemma the best action for suspect 1 when suspect 2 finks is fink and also when suspect 2 chooses quiet is fink.

PROPOSITION: The **action profile** $a^* = \{a_1^*, \dots, a_n^*\}$ is a Nash equilibrium of a strategic game with ordinal preferences if and only if every player's action is a best response to the other players' actions. Formally: $a_i^* \in B_i(a_{-i}^*)$ for every player i .

The above proposition provides a simple algorithm to find a Nash equilibrium that reduces the number of player preferences to examine.

Algorithm for finding Nash equilibria:

1. find the best response function of each player
2. find the action profiles $a_i^* \in B_i(a_{-i}^*)$ for every player i .

Variant of Stag Hunt

Consider the following variant of n -hunter Stag Hunt:

- only m hunters, with $2 \leq m < n$, need to pursue the stag in order to catch it.
- there is still a single stag
- stag is shared only by the hunters that catch it
- that each hunter prefers the fraction $1/n$ of the stag to a hare.

Let's see as for example the case with $n=4$ and $m=2$ but the same considerations apply for every value of n .

Hunter 1	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>
Hunter 2	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>	<u>H</u>
Hunter 3	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>S</u>	<u>S</u>	<u>H</u>	<u>H</u>	<u>H</u>
Hunter 4	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>S</u>	<u>H</u>	<u>H</u>

We highlight in red the best response for each player. In yellow the two Nash equilibria. Observe that using the algorithm for finding Nash equilibrium using best response we don't need to

evaluate all the players' preferences but only the subset in common among the already observed players (Underlined elements)

Homework: consider a further variation of the stag hare problem where the hunter prefers $1/k$ of the stag to the hare.